

Kyung Hyun Lee

AI Researcher in Physiological and Clinical Time-Series Learning

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Summary

AI researcher developing and evaluating machine learning models for physiological signals and longitudinal clinical data, with particular attention to external validation and continuous clinical monitoring. PhD student in Digital Health at SAIHST, Sungkyunkwan University, and AI researcher at AITRICS. Co-founder & CEO of BreathYou Co., Ltd.

Education

PhD Student	Sungkyunkwan University (SAIHST) , Digital Health • Advisor: Prof. Byung-Jae Lee • Research on Transformer-based trajectory models of respiratory health	Seoul, South Korea 2026 – present
Master of Science	Sungkyunkwan University (SAIHST) , Digital Health • Advisor: Prof. Mira Kang • M Lab. — Statistical analysis and predictive model development using medical big data	Seoul, South Korea 2019 – 2021
Bachelor of Science	Incheon National University , Electronic Engineering & Computational Neuroscience • Advisor: Prof. Sangwon Byun • Bioelectronics Lab. — Biosignal (ECG, EMG) data collection, signal analysis, and real-time prediction model development	Incheon, South Korea 2014 – 2019

Experience

University of Seoul , Lecturer Faculty appointment in the Department of Sport Science, teaching the undergraduate course Exercise and Digital Healthcare. • Designed and teach Exercise and Digital Healthcare, a fourth-year major elective (Fall 2026)	Seoul, South Korea 2026 – present 1 year
BreathYou Co., Ltd. , Co-founder & CEO Digital health startup focused on allergy and respiratory AI, developing DeepBreath CDSS and MediPipe. • Selected for 2025 Pre-startup Package • Incorporated on 5 August 2026 • DeepBreath CDSS: Clinical decision support for predicting pulmonary function decline trajectories • MediPipe: OCR/NLP pipeline for medical record processing	South Korea 2025 – present 1 year

AITRICS , Researcher (2024 – present) · Industrial Technical Personnel — Alternative Military Service (2022 – 2024) Medical data (RWD) cleaning, analysis pipeline construction, predictive model development, and external validation. Contributed to the cardiac arrest early-warning model that obtained MFDS (Korean FDA) approval as an AI medical device, and currently leading product development of a self-developed AKI prediction model toward MFDS approval. • Research design for clinical data analysis projects • Data extraction and cleaning from multi-center hospitals • Automated data analysis pipelines using Airflow • Deep learning-based clinical time-series prediction models • Clinical simulation-based model evaluation and survival analysis • Cardiac arrest prediction model — supported MFDS approval as an AI medical device • AKI prediction model — developed the model and currently leading productization toward MFDS approval	Seoul, South Korea 2022 – present 4 years
Exosystems , Biosignal AI Research Engineer Deep learning-based EMG modeling for Muscle Quality monitoring. • RNN, CRNN-based Muscle Quality time-series prediction model development • Clinical validation-oriented AI model evaluation	South Korea May 2022 – Aug 2022 4 months
Incheon National University , Researcher • EMG-based real-time hand gesture recognition algorithm R&D • Deep learning-based ECG modeling for autonomic nervous system monitoring	Incheon, South Korea Sept 2021 – May 2022 9 months
Samsung Medical Center & SAIHST , Researcher EHR data extraction and analysis from CDW; development of ML-based CDSS. • Data cleaning and health checkup database construction • Text mining of unstructured medical records • Statistical analysis and predictive/diagnostic model development	Seoul, South Korea Dec 2018 – Aug 2021 2 years 9 months

Teaching

University of Seoul, Department of Sport Science: Lecturer · Exercise and Digital Healthcare, fourth-year undergraduate elective (Fall 2026)

University of Seoul, Department of Sport Science: Invited seminar · Healthcare industry, two hours, hosted by Prof. Sae Young Jae (2024)

Incheon National University: Guest lecture · Student-Designed Convergence Major, one hour (2024)

Awards

Lab Startup Competition — Excellence Award Sungkyunkwan University, Start-up Support Foundation • University-wide competition among Sungkyunkwan University graduate-student laboratory startup clubs • MediPipe: on-premise medical data infrastructure software that turns in-hospital test results and clinical records into AI-ready data	2026
Consortium Lab-based Startup Camp — Excellence Award Sungkyunkwan University, Startup-Centered University program • Consortium competition among laboratory startup teams from universities across the Seoul metropolitan area • MediPipe: the data infrastructure that lets hospitals use AI	2026
Human Understanding AI Paper Competition — Top 7 & ETRI Director's Award ETRI, Ministry of Science and ICT • Multimodal Attention-based Korean emotion recognition model combining RoBERTa (Text) and Wav2vec 2.0 (Audio)	2022

OCR AI Training Data Hackathon — Excellence Award Ministry of Science and ICT, NIA AI-Assisted OCR Annotation System for pharmaceutical products (CRAFT-based)	2021
Medical Information Analysis Expert Advanced Course — Grand Prize Ministry of Health and Welfare, Asan Medical Center Real-time BIS prediction model using intraoperative EEG data (25-second lookahead)	2021
Medical AI Expert Training Course — Excellence Award Korea Human Resource Development Institute for Health & Welfare 30-day mortality and ICU admission prediction model using preoperative and intra-operative vital signs	2020
STAR Program Paper Presentation — Grand Prize DSP-IN-U Smart Device Talent Development Program EMG-based real-time hand gesture prediction model development	2018

Academic Service

Journal Reviewer: Computers in Biology and Medicine (4), Frontiers in Medicine (2), Frontiers in Digital Health, Frontiers in Network Physiology, Frontiers in Psychiatry, Information Processing & Management · 2026–present

Conference Reviewer: NeurIPS Workshop: Responsible Communication of Machine Learning Research in Biomedicine (RCMLR) · 2026

Certificates

Medical Information Analysis Expert Advanced Course Korea Human Resource Development Institute for Health & Welfare, Asan Medical Center	Aug 2021
Google Developer Machine Learning Boot Camp Google Developers	Jan 2021
Medical AI Expert Training Course Korea Human Resource Development Institute for Health & Welfare	Dec 2020

Skills

Programming & ML: Python, PyTorch, TensorFlow, Scikit-learn

Data & Database: SQL (PostgreSQL), Airflow, Data Pipeline

Statistics & Analysis: R, Survival Analysis, Clinical Statistics

Web Development: FastAPI, Flask, JavaScript, HTML/CSS, AWS

Languages

Korean

Native speaker

English

Professional working proficiency

References

Prof. Sangwon Byun

Incheon National University, Department of Electronic Engineering; Bachelor's advisor
([Bioelectronics Lab.](#) · [Scholar](#))

Prof. Mira Kang

Samsung Medical Center; Master's advisor at SAIHST, Sungkyunkwan University

Prof. Jin-Young Lee

Samsung Medical Center, Health Promotion Center

Prof. Byung-Jae Lee

Samsung Medical Center, Department of Allergy and Clinical Immunology; Doctoral advisor at SAIHST, Sungkyunkwan University

Prof. Sae Young Jae

University of Seoul, Department of Sport Science; research collaborator